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# HAPPINESS DATA ANALYSIS

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# Research Background

Happiness is often understood as something personal and subjective. However, in modern statistical studies, happiness can be measured and analyzed through social, economic, and quality-of-life indicators of a country. This research aims to understand the factors that significantly influence the happiness levels of countries around the world and to examine how the distribution of happiness varies across regions.

The dataset used comes from the Kaggle platform, titled "World Happiness Cleaned Dataset," which is a version of the global happiness data that has undergone a cleaning and standardization process. This dataset is retrieved from the link: "https://www.kaggle.com/datasets/seanlim123/world-happiness-cleaned".

This data contains happiness scores and supporting variables such as economics (GDP per capita), social support (Family), health, freedom, trust in government, and generosity. Using this dataset, the analysis was conducted to understand global happiness patterns in a more objective and measurable way.

## Analysis Methodology

This research uses a quantitative approach with a combination of descriptive and inferential statistics. The analysis flow is structured systematically as follows:

1. Exploratory Data Analysis (EDA)

The data is cleaned and only relevant numeric variables are selected. This stage aims to understand the initial structure of the data, detect anomalies, and ensure readiness for further analysis.

2. Pearson Correlation Test

Used to measure the strength of the linear relationship between numerical variables and the Happiness Score. The results are visualized in a heatmap to facilitate interpretation of the relationship patterns.

3. Simple Linear Regression

Performed using a scatter plot with an OLS trendline to examine the individual influence of each variable on the happiness score.

4. Multiple Linear Regression

A model was constructed using six main variables as predictors. Evaluation was carried out using $R^2$, F-test, coefficient significance test, and testing of model assumptions such as multicollinearity (VIF) and heteroscedasticity (Breusch-Pagan).

5. Predictive Model Evaluation
The model was tested using RMSE, $R^2$ Score, and MAPE metrics to measure its predictive performance against actual data.

6. Chi-Square Test
Used to test whether there is a significant relationship between happiness level categories (Low, Medium, High) and geographic region (Region).

This method allows for a comprehensive analysis: not only examining correlations and predictions, but also understanding the structure of social relationships in happiness in a more in-depth and measurable way.

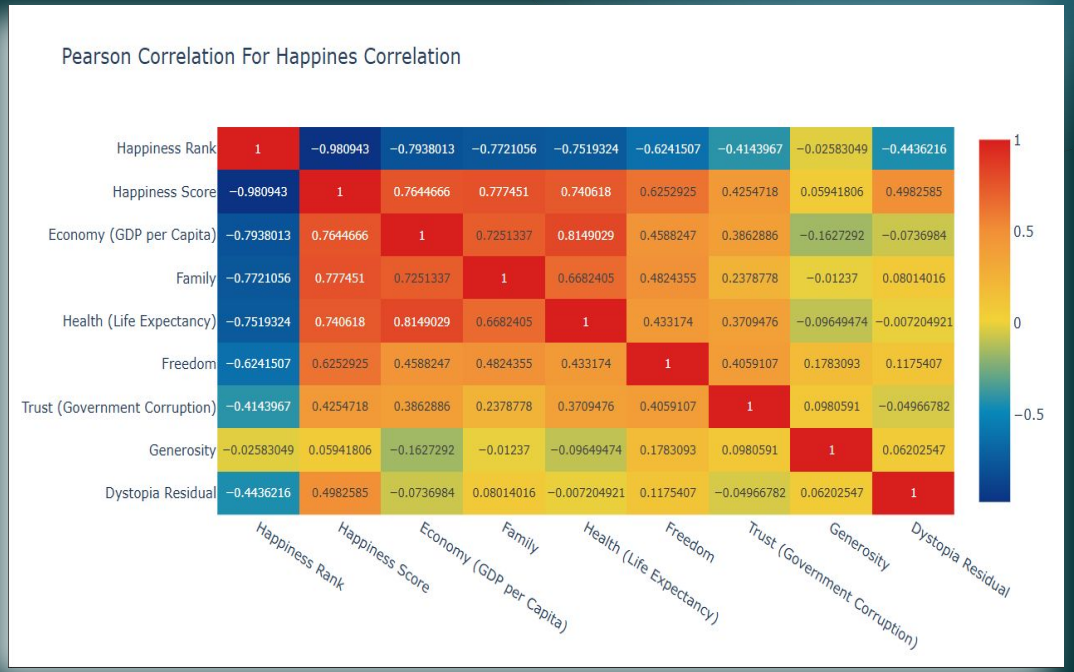
# PEARSON CORRELATION

INTERPRETATION :

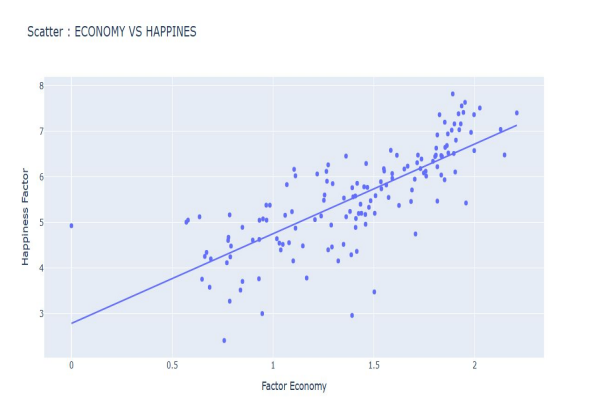
1. “A country’s happiness level is more influenced by basic human factors: economic stability, health, and social support, than by moral factors like generosity.”

2. “Personal freedom and trust in government also contribute, but not as strongly as economic factors and health.”

3. “The happiness ranking is inversely proportional to the happiness score, thus indicating structural consistency in the data.”



# TRENDLINE


Scatter : ECONOMY VS HAPPINES
Happiness Factor
Factor Economy

FAMILY VS HAPPINESS SCORE
Happiness Factor
Family (SOcial Support)

Generosity VS Happiness Factor
Happiness Factor
Generosity Factor

Health (Life Expectancy) VS Happiness Score
Happiness Factor
Health (Life Expectancy)

Freedom VS Happiness Score
Happiness Factor
Freedom Factor

# INTERPRETATION OF THE OLS TRENDLINE :

1. There is a strong positive relationship between Family (Social Support) and Happiness Score. This means that the higher a country's social support, the higher its happiness level.

2. The slope = 3.03. This means: If the Social Support value increases by 1 unit, the Happiness Score tends to increase by about 3 points. This is a very large effect, almost 3x stronger than the economic effect I found earlier (around 1.97).

3. $R^2$ = 0.604. This means: 60.4% of the variation in Happiness Score can be explained by Social Support alone. This is the largest contribution compared to economics, health, freedom, generosity, trust, etc.

In Conclusion :

Social support is the most basic human need—feeling safe, having someone to help during a crisis, and it turns out to be a stronger driver of happiness rather than money, freedom, or health. Social Support is the strongest individual predictor of happiness in this dataset according with statistical method.

# MULTIPLE LINEAR REGRESSION

The linear regression model used demonstrated excellent performance in explaining variations in happiness levels. The R-squared value of 0.773 indicates that 77.3% of the variation in the Happiness Score can be explained by the six analyzed variables. After adjusting for the number of variables, the Adjusted R-squared remained high (0.763), indicating the model is stable and does not experience overfitting. The F-test also showed highly significant results (p-value $\ll$ 0.05), indicating that overall, this combination of variables is effective in predicting happiness levels.

Partially, the strongest and most significant factor is Family (social support), with a coefficient of 1.40, meaning a 1-unit increase in social support increases the Happiness Score by approximately 1.40 points. This finding is consistent with the previous scatterplot results. Other significant factors are Freedom (1.60) and Health (1.27), indicating that individual freedom and health conditions contribute significantly to happiness. Economic factors also have a significant effect, although their effect is smaller (0.55).

Conversely, Trust in Government and Generosity show a positive influence, but are not statistically significant after controlling for other variables. This indicates that these two factors are not strong enough to be the main predictors of happiness in this model.

**Key takeaway :**

**Happiness is more determined by the quality of social relationships, personal freedom, and health than by trust in government or levels of generosity.**

# INTERPRETATION FOR MULTICOLINEARITY :

The results of the multicollinearity check indicate that all variables have VIF values below 5, indicating no multicollinearity issues in the model. This means that each variable operates independently, does not replicate each other, and there is no extreme overlap between predictors. Therefore, the regression model used is stable and reliable, as each variable provides relevant new information. In detail:

1. GDP per Capita (VIF = 3.96)

   Slightly correlated with other variables such as Health or Family, but still within safe limits. This variable still makes a unique contribution to the happiness score.

2. Family/Social Support (VIF = 2.40)

   The relationship with other variables is moderate and does not cause conflict. This variable stands alone and does not overlap with GDP or Health.

3. Health (VIF = 3.14)

   There is a natural relationship with GDP because more developed countries tend to have better health, but its value remains stable and does not confound the model.

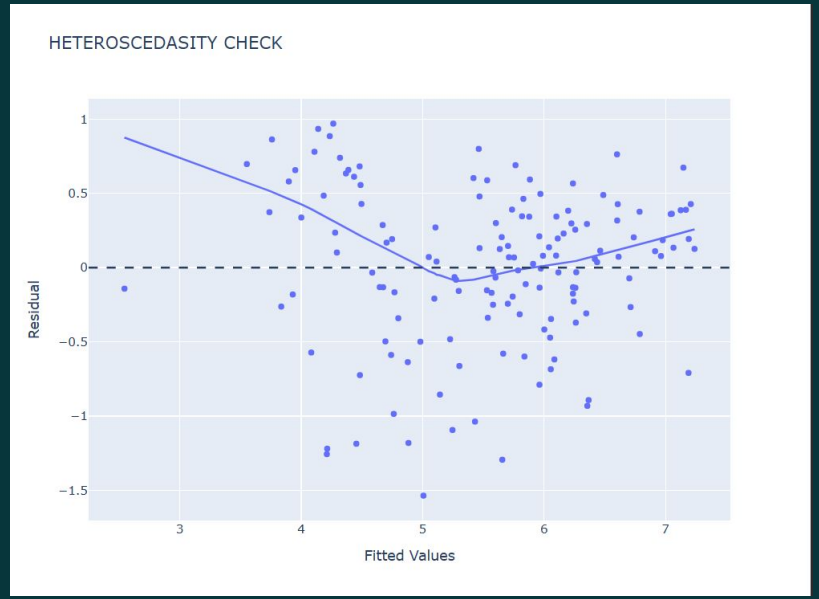
4. Freedom, Trust, and Generosity (VIF 1.1–1.5)

   There is virtually no correlation between the variables—they are the cleanest predictors and provide completely independent additional information.

**Overall, the model does not suffer from multicollinearity issues, so all variables can be used together in the regression without the need to drop any variables.**

# INTERPRETATION HETEROSCEDASTICITY WITH VISUALIZATION :

"The plot of residuals vs. fitted values shows a clear pattern. The LOWESS line is not flat around zero, but forms a curved pattern like an 'inverted mountain'. This indicates that the residual variance is not constant. In other words, the model suffers from heteroscedasticity, because the error is larger at some levels of prediction than at others. This pattern indicates that the model is not yet fully stable and its standard estimates may be biased."



## INTERPRETATION HETEROSCEDASTICITY :

The Breusch-Pagan test results showed a BP value of 22.91 with a p-value of 0.00083, which is below 0.05. This indicates that the model experiences heteroscedasticity, a condition where the error variance is not constant, thus potentially biasing the standard error in a regular OLS model. To maintain model validity, I then applied a correction using heteroscedasticity-robust standard errors (HC3). After the correction, the regression coefficient value remained unchanged, the R-squared remained the same, and the F-statistic remained significant, but the standard error has been adjusted to improve the estimation accuracy. Thus, even though heteroscedasticity was detected, the model can still be used safely because all coefficient interpretations, p-values, and significance have been corrected using the appropriate method.

## INTERPRETATION FOR ANALZE BASED ON THE MODEL :

1. The constructed multiple linear regression model explained 77% of the variation in Happiness Score values based on the variables of economics, family, health, freedom, trust in government, and generosity.

2. The model's prediction error rate was relatively low, with an RMSE of 0.52, meaning the average difference between actual and predicted values was only about half a point on the happiness scale.

3. Furthermore, the MAPE value of 8.3% indicates that the model's relative error is quite small, indicating that the model performs well and is suitable for use in predictive happiness analysis on the 2022 dataset.

# CHI-SQUARE TESTING

# CHI SQUARE INTERPRETATION :

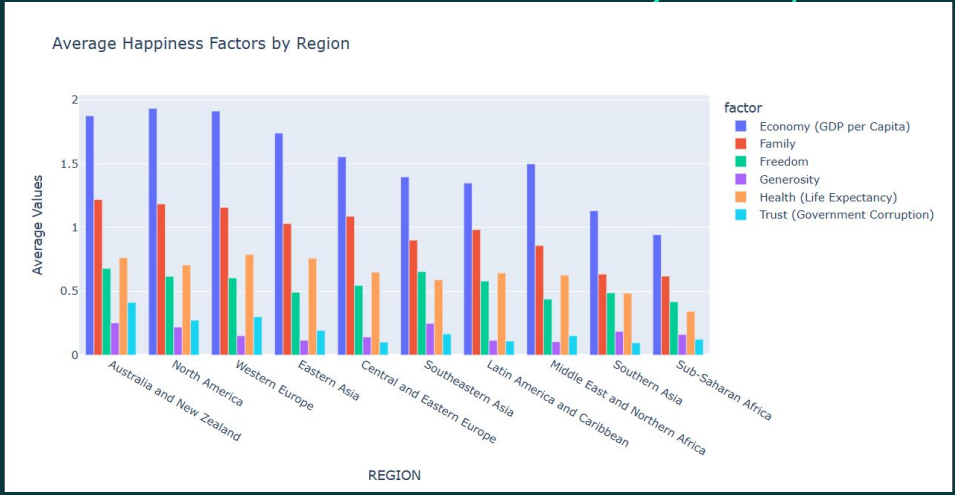
Based on the Chi-Square test, the p-value obtained was $1.69 \times 10^{-16}$, which is much smaller than $\alpha = 0.05$. This indicates a significant relationship between Region and happiness level categories. Thus, the null hypothesis is rejected, meaning the distribution of happiness levels does not occur randomly across regions, but is influenced by regional factors.

# DESCRIPTIVE STATISTICS TO GIVE AN OVERVIEW

# INSIGHT AVERAGE HAPPINESS FACTOR BASED ON REGION :

Conversely, regions such as Sub-Saharan Africa and Southern Asia show low scores in almost all factors, particularly Economy, Trust, and Health, indicating structural limitations and weak life support systems. Interestingly, the Generosity and Freedom factors are not always high in economically strong regions, indicating that happiness is not simply a matter of material things, but also the quality of relationships and social freedom.

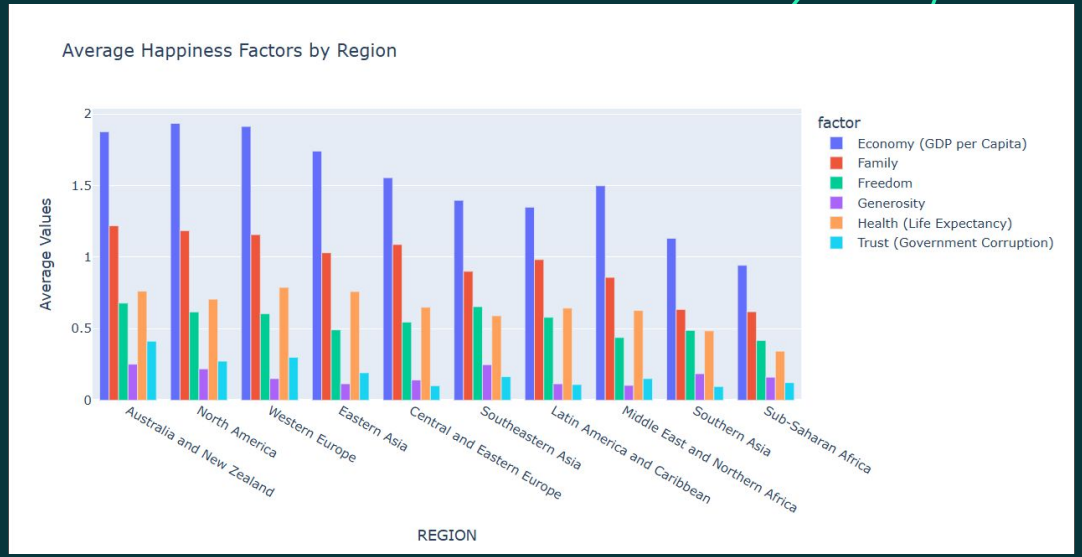
This reinforces the notion that happiness in developed regions tends to be "stable but fragile" because it is highly dependent on the stability of economic and institutional systems.




The visualization shows that the Economy (GDP per capita) factor is the most dominant determinant in almost all regions, particularly Australia & New Zealand, North America, and Western Europe. This demonstrates that economic well-being remains the primary foundation of measurable happiness. This is followed by Family and Health factors, which are relatively stable in developed regions, indicating social support and a more secure quality of life.

# RECOMMENDATION AVERAGE HAPPINESS FACTOR BASED ON REGION :

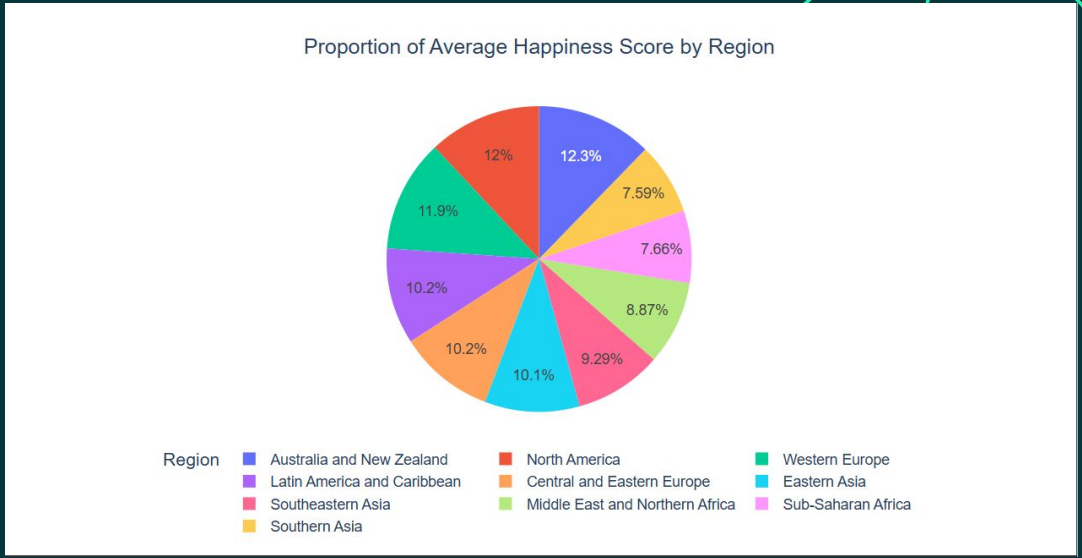
Referring to Erich Fromm's thoughts in *To Have or To Be?*, this data demonstrates the need for a paradigm shift in development from a focus on "having" to "being." If happiness continues to be reduced solely to economic figures, society risks becoming trapped in an existentially fragile illusion of well-being.




Therefore, the recommendation is to encourage social policies and approaches that not only increase income but also strengthen the meaning of life, social solidarity, and the quality of human relationships. This way, happiness will not only be statistically high but also more authentic, deeply rooted, and sustainable from a human perspective.

# INTERPRETATION BASED ON THE PIE CHART I CREATED :

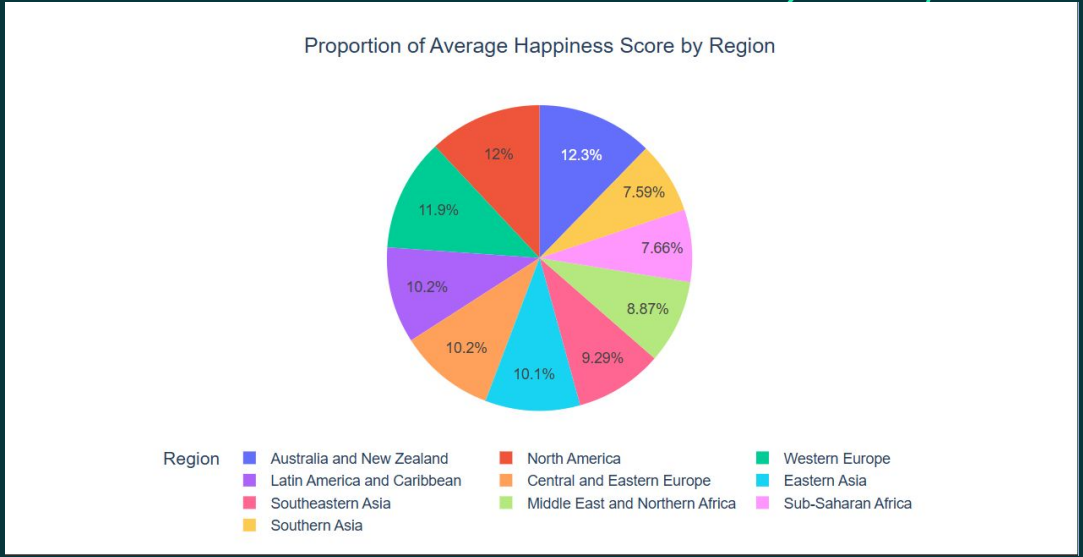
The graph shows that global happiness is not evenly distributed. Australia & New Zealand, North America, and Western Europe dominate the happiness score, indicating that economic stability, quality of public services, and social security play a significant role in shaping a consistent quality of life.




Conversely, Sub-Saharan Africa and Southern Asia have the lowest proportions, indicating structural limitations such as poor access to basic necessities and weak social resilience. Although developed regions excel, happiness there tends to be "stable but fragile" in the face of economic and social crises, as it is highly dependent on the sustainability of the systems that support it.

# RECOMMENDATION BASED ON THE PIE CHART I CREATED :

Referring to Erich Fromm, happiness should not be pursued solely through material improvements and statistical indicators, but through a shift from an orientation toward "having" to "being." This means that development should not only target economic growth but also strengthen the quality of human relationships, a sense of meaning, and a sense of life.




With this approach, regions with high scores can deepen the quality of their happiness beyond mere numerical stability, while regions with low scores can build a more humane and sustainable foundation for happiness.

# FINAL CONCLUSION AND REFLECTION :

Based on the statistical analysis conducted that from data exploration, correlation, chi-square tests, to linear regression, it appears that a country's happiness level is influenced by a combination of social, economic, and health factors. Social support emerged as the factor most strongly associated with the Happiness Score, followed by GDP per capita and health.

The chi-square analysis also showed that the distribution of happiness levels differed significantly across regions, indicating that geographic context also shapes a community's socio-economic conditions. The regression model constructed was able to explain most of the variation in happiness scores and remained valid despite heteroscedasticity, as it was corrected using robust standard errors. Overall, the data show that countries with strong social support, stable economies, and good health tend to have higher levels of happiness.

# REFLECTION :

EDA and preprocessing are important because they help ensure that the data is fit for analysis. This process detects issues such as missing values, outliers, incorrect data types, or inappropriate distributions, which can compromise the validity of statistical analysis. If data is not cleaned, correlation, chi-square, and regression results can be misleading due to noise or errors. By performing EDA first, we improve the quality of insights and ensure the statistical models we build are more accurate, stable, and reliable.

In the professional world, these three analyses are frequently used. Correlation analysis is used to find relationships between indicators, for example, in customer analytics to see which factors are associated with increased retention. Chi-square analysis is used in categorical data, for example, to see whether product preferences differ across regions. Linear regression is widely used for forecasting or predicting business performance, such as sales, customer lifetime value, or revenue. These analyses help companies make decisions based on data, not assumptions.

# GOOGLE COLLAB LINK :

https://colab.research.google.com/drive/1diahQvGfBgJ6_YrGvzUvSjX0e6jmwnYf?usp=sharing

# DATA SCOURCE LINK :

"https://www.kaggle.com/datasets/seanlim123/world-happiness-cleaned".

# Thank you!